

Linear modeling techniques for liquid height regulation in two tank - Two variable system

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ABSTRACT

In this paper, two linear modeling techniques are presented for regulating the liquid height in two tank - two variable system (TT-TVS). Initially, the nonlinear dynamics of TT-TVS are linearized using the Taylor series linearization technique. However, this method reveals that some parameters of TT-TVS are inexact, necessitating the adoption of system identification techniques or mathematical approaches to accurately identify the linear dynamics. To address this, two new approaches are proposed: (i) a mathematical approach utilizing real-time input-output data, and (ii) a Linear Variable Parameter Transfer Function (LVPTF) model identification approach, which employs real-time data and MATLAB curve fitting tool. A comparative analysis between the proposed identification techniques and existing methods from the literature is also presented. The results indicate that the LVPTF modeling technique offers superior accuracy in identifying the linear dynamics of TT-TVS.

1. Introduction

There are many foundries such as drinking water treatment foundry or RO foundry [1], chemical foundry [2], food handling foundry [3], Nuclear electricity generation stations [4], boilers in thermal power stations [5], etc have two tank - two variable system (TT-TVS). In 90s, International Federation of Automatic Control (IFAC) advisory committee has notified a set of real time control problems that are used to test the performance of newly designed control algorithms and control tools. The advisory committee notified a set of real time control problems that are named as 'real time control benchmark problems'. Out of which, the liquid height control problem in TT-TVS is featured as a benchmark problem in implementing two variable control algorithms [6]. Since last twenty years, liquid height control in TT-TVS has attracted the focus of researchers. It is one of the most challenging benchmark control problem to implement MIMO control algorithms.

All real time variable regulation problems in real time systems start with notifying the dynamics of the real time systems. To notify the dynamics of real time systems, system identification techniques have been used [6]. System identification durably built the relation between the real time systems and notified dynamics of the real time systems. System identification has its early roots in Eighteenth and Nineteenth century introduced in Refs. [7,8]. The first landmark invention in the estimation

of system parameters was performed with Kalman filter in 1961 [9]. Second remarkable invention towards the system identification was made by Cooley and Tukey [10]. [10] introduced Fast Fourier Transform (FFT) algorithm for signal spectral analysis. In mid years of 1970s, system identification was transformed to another level with the introduction of prediction error minimization (PEM) [6]. However, from the literature [6,11] survey, it is observed that Grey-box and Black-box identification is proposed using MATLAB system notifying mechanism to notify the linear dynamics of coupled tank system. Linear Single-Input Single-Output (SISO), Multiple-Input Single-Output (MISO), and MIMO models of coupled tank system were identified in the form OE, ARX, ARMAX, and state-space model structures. In Ref. [12], subspace algorithm N4SID is chronicled to notify the states of in the state-space form. In Ref. [13], least square support vector regression method based on genetic optimization algorithm is proposed to identify the dynamics of Aircraft engine. In Ref. [14], a piece wise two layer model is proposed for model predictive control. A new system identification method is proposed for boiler drum in Ref. [15].

Recently, two tank - two variable system identification area, linear dynamics identification methods are proposed for two tank - two variable system in Refs. [16,17]. As per the best knowledge of literature survey [6,11,12,16,17], it is observed that most of the modeling techniques to model the coupled tank system are system identification

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techniques but no mathematical approach is proposed. In the present work, (i) A novel mathematical approach is proposed to identify the dynamic of TT-TVS. (ii) A novel linear variable parameter transfer function (LVPTF) model identification is recommended to notify the dynamics of TT-TVS which is not only designer friendly but also provides scope for adaptive controller design.

2. Description of TT-TVS

The real time layout of TT-TVS is shown in Fig. 1.

It has four identical, cylindrical shaped primary tanks and a reservoir tank. Each primary tank consists of level detecting sensor, inlet and outlet. Two DC motor pumps are available in the reservoir tank. To model the TT-TVS, the voltage signals given to the merged motor pumps are taken as input parameters and water heights in the tanks as output parameters. One interface card is available to get the output signals of sensors and to send the control signal to actuators. The data set collected from the real time experiment can be analyzed in MATLAB/Simulink environment. In the present work, upper two tanks, i.e., Tank-1 and Tank-2 are coupled with each other through the coupling path. Therefore, the dynamics of the upper two tanks can be modeled as a Two-Input Two-Output (TITO) plant. The nonlinear relations between the outputs and inputs of TT-TVS are given as [6]:

$$\left. \begin{aligned} \frac{dh_1(t)}{dt} &= -\frac{a_1}{A} \sqrt{2gh_1(t)} - \frac{a_{12}}{A} \sqrt{2gh_1(t) - 2gh_2(t)} + \eta u_1(t) \\ \frac{dh_2(t)}{dt} &= -\frac{a_2}{A} \sqrt{2gh_2(t)} + \frac{a_{12}}{A} \sqrt{2gh_1(t) - 2gh_2(t)} + \eta u_2(t) \end{aligned} \right\} \quad (1)$$

where h_j is the liquid level in j th Tank, $j \in (1, 2)$; u_j is the voltage given to j th DC motor pump; a_j is the outgoing pipe area of j th Tank; a_{12} is the cross-sectional area of the pipe from Tank-1 to Tank-2; A is the area of the tanks, η relates the control input with water inflow from the pumps; g is the gravitational constant (9.81 m/s^2). However, from the literature [6,18,19], it is observed that a linear transfer function model is preferable for designing a controller using both frequency domain specifications and time domain specifications. This preference arises because linear transfer function models provide a clear and systematic framework for analyzing system behavior and performance. In the frequency domain, these models allow for precise tuning of controllers to achieve desired specifications and stability margins. Similarly, in the time domain, they facilitate the accurate prediction and adjustment of transient response characteristics. Consequently, linear transfer function models offer a robust and versatile approach to controller design, ensuring that both frequency and time domain requirements are met effectively.

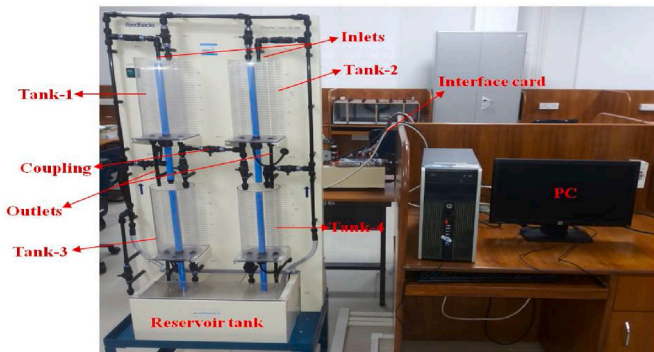


Fig. 1. Real time diagram of TT-TVS.

3. Linearization of TT-TVS dynamics using Taylor series linear technique

In the present section, the non-linear equations given in (1) are linearized by applying the Taylor series linearization technique. This process involves linearizing the equations around the steady-state operating points, denoted as $\bar{h}_1$ and $\bar{h}_2$ with respect to the variables h_1 and h_2 . The linearization procedure simplifies the dynamics described in (1), making the equations more manageable and easier to analyze. The detailed steps of this procedure are as follows: the dynamics provided in (1) are transformed and approximated using Taylor series expansion around the steady-state points, resulting in a set of linear equations that accurately represent the system behavior in the vicinity of these operating points. The dynamics given in (1) are simplified as:

$$\frac{dh_1(t)}{dt} = -f_1 - f_{12} + \eta u_1(t) \quad (2)$$

$$\frac{dh_2(t)}{dt} = -f_2 + f_{12} + \eta u_2(t) \quad (3)$$

By comparing (1) and (2),

$$f_1 = \left(\frac{a_1}{A} \sqrt{2g} \right) \sqrt{h_1(t)} = k_1 \sqrt{h_1(t)} \quad (4)$$

$$f_2 = \left(\frac{a_2}{A} \sqrt{2g} \right) \sqrt{h_2(t)} = k_2 \sqrt{h_2(t)} \quad (5)$$

$$f_{12} = \left(\frac{a_{12}}{A} \sqrt{2g} \right) \sqrt{(h_1(t)) - (h_2(t))} = k_{12} \sqrt{(h_1(t)) - (h_2(t))} \quad (6)$$

According to Taylor series expansion, if a function $f(x)$ is continuous and differentiable at a point $x = x_0$, then $f(x)$ can be expressed as:

$$f(x) = f(x_0) + \left| \frac{df}{dx} \right|_{ss} \frac{(x - x_0)}{1!} + \frac{d^2f}{dx^2} \frac{(x - x_0)^2}{2!} + \dots \quad (7)$$

where 'ss' refers steady-state operating point, i.e., at $x = x_0$. By applying Taylor series expansion, $f_{12} = k_{12} \sqrt{(h_1(t)) - (h_2(t))}$ is expressed as:

$$f_{12} = \bar{f}_{12} + \left| \frac{df_{12}}{dh_1} \right|_{ss} (h_1(t) - \bar{h}_1) + \left| \frac{df_{12}}{dh_2} \right|_{ss} (h_2(t) - \bar{h}_2) \quad (8)$$

From (8), one obtains

$$f_{12} = \bar{f}_{12} + m_{12}[(h_1(t) - \bar{h}_1) - (h_2(t) - \bar{h}_2)] \quad (9)$$

In (9),

$$m_{12} = \frac{k_{12}}{2}(\bar{h}_1 - \bar{h}_2)^{-\frac{1}{2}} \quad (10)$$

By applying Taylor series expansion, $f_1 = k_1 \sqrt{h_1(t)}$ is expressed as:

$$f_1 = \bar{f}_1 + \left| \frac{df_1}{dh_1} \right|_{ss} (h_1(t) - \bar{h}_1) \quad (11)$$

From (11), one obtains

$$f_1 = \bar{f}_1 + m_1(h_1(t) - \bar{h}_1), \quad (12)$$

In (12),

$$m_1 = \frac{k_1}{2}(\bar{h}_1)^{-\frac{1}{2}} \quad (13)$$

By applying Taylor series expansion method, $f_2 = k_2 \sqrt{h_2(t)}$ is expressed as:

$$f_2 = \bar{f}_2 + \left| \frac{df_2}{dh_2} \right|_{ss} (h_2(t) - \bar{h}_2) \quad (14)$$

From (14), one obtains

$$f_2 = \bar{f}_2 + m_2(h_2(t) - \bar{h}_2) \quad (15)$$

In (15),

$$m_2 = \frac{k_2}{2}(\bar{h}_2)^{-\frac{1}{2}} \quad (16)$$

By substituting (8) and (11) in (2), one obtains

$$\begin{aligned} \frac{dh_1(t)}{dt} = & -\bar{f}_1 - \bar{f}_{12} - (m_1 + m_{12})(h_1(t) - \bar{h}_1) \\ & + m_{12}(h_2(t) - \bar{h}_2) + \eta u_1(t) \end{aligned} \quad (17)$$

At steady-state $\frac{dh_1(t)}{dt} = 0$. Due to this, at $\bar{h}_1$, (2) can be derived as

$$\frac{d\bar{h}_1}{dt} = -\bar{f}_1 - \bar{f}_{12} + \eta \bar{u}_1 = 0 \quad (18)$$

Subtracting (18) from (17), one obtains

$$\frac{d\hat{H}_1(t)}{dt} = m_{12}\hat{H}_2(t) - (m_1 + m_{12})\hat{H}_1(t) + \eta\hat{U}_1(t) \quad (19)$$

Similarly, using (3) and Taylor series expansion method, the linearized dynamics for Tank-2 are obtained as:

$$\frac{d\hat{H}_2(t)}{dt} = m_{12}\hat{H}_1(t) - (m_2 + m_{12})\hat{H}_2(t) + \eta\hat{U}_2(t) \quad (20)$$

where $\hat{H}_1(t)$, $\hat{H}_2(t)$, $\hat{U}_1(t)$ and $\hat{U}_2(t)$ are the deviation variables of $h_1(t)$, $h_2(t)$, $u_1(t)$ and $u_2(t)$ respectively and given as:

$$\begin{cases} \hat{H}_1(t) = h_1(t) - \bar{h}_1 \\ \hat{H}_2(t) = h_2(t) - \bar{h}_2 \\ \hat{U}_1(t) = u_1(t) - \bar{u}_1 \\ \hat{U}_2(t) = u_2(t) - \bar{u}_2 \end{cases} \quad (21)$$

Taking s-domain transform of (19), one obtains

$$\begin{aligned} \hat{H}_1(s) = & \left(\frac{m_{12}}{s + (m_1 + m_{12})} \right) \hat{H}_2(s) \\ & + \left(\frac{\eta}{s + (m_1 + m_{12})} \right) \hat{U}_1(s) \end{aligned} \quad (22)$$

Taking s-domain transform of (20), one obtains

$$\begin{aligned} \hat{H}_2(s) = & \left(\frac{m_{12}}{s + (m_2 + m_{12})} \right) \hat{H}_1(s) \\ & + \left(\frac{\eta}{s + (m_2 + m_{12})} \right) \hat{U}_2(s) \end{aligned} \quad (23)$$

By substituting (23) in (22) and (22) in (23), linear dynamics of TT-TVS are obtained as:

$$\begin{cases} \hat{H}_1(s) = G_{11}(s)\hat{U}_1(s) + G_{12}(s)\hat{U}_2(s) \\ \hat{H}_2(s) = G_{21}(s)\hat{U}_1(s) + G_{22}(s)\hat{U}_2(s) \end{cases} \quad (24)$$

In (24), $G_{11}(s)$, $G_{12}(s)$, $G_{21}(s)$ and $G_{22}(s)$ are as follows:

$$\begin{aligned} G_{11}(s) = & \left. \begin{aligned} & \frac{\hat{H}_1(s)}{\hat{U}_1(s) \mid \hat{U}_2(s)=0} \\ & = \frac{\eta[s + (m_2 + m_{12})]}{[s + (m_1 + m_{12})][s + (m_2 + m_{12})] - m_{12}^2} \end{aligned} \right\} \quad (25) \end{aligned}$$

$$\begin{aligned} G_{12}(s) = & \left. \begin{aligned} & \frac{\hat{H}_1(s)}{\hat{U}_2(s) \mid \hat{U}_1(s)=0} \\ & = \frac{\eta \times m_{12}}{[s + (m_1 + m_{12})][s + (m_2 + m_{12})] - m_{12}^2} \end{aligned} \right\} \quad (26) \end{aligned}$$

$$G_{21}(s) = \left. \begin{aligned} & \frac{\hat{H}_2(s)}{\hat{U}_1(s) \mid \hat{U}_2(s)=0} \\ & = \frac{\eta \times m_{12}}{[s + (m_1 + m_{12})][s + (m_2 + m_{12})] - m_{12}^2} \end{aligned} \right\} \quad (27)$$

$$G_{22}(s) = \left. \begin{aligned} & \frac{\hat{H}_2(s)}{\hat{U}_2(s) \mid \hat{U}_1(s)=0} \\ & = \frac{\eta[s + (m_1 + m_{12})]}{[s + (m_1 + m_{12})][s + (m_2 + m_{12})] - m_{12}^2} \end{aligned} \right\} \quad (28)$$

From (24), it is noted that parameters m_1 , m_2 , m_{12} and η are inexact. Furthermore, (1) indicates that un-modeled dynamics such as sloshing, turbulence, and moisture reactions are not accounted for in the dynamics of the TT-TVS. Therefore, there is a need for adaptive system identification techniques or mathematical approaches to accurately identify the linear dynamics of the TT-TVS. In this linear dynamics identification process, the definition and use of deviation variables in (21) is crucial. According to the linearization method described above and (21), it can be observed that the water height in Tank-1 is linearized at the operating $\bar{h}_1$, where $\bar{h}_1$ is the initial steady-state water height for the corresponding initial input $\bar{u}_1$. Similarly, the water height in Tank-2 is linearized at the operating point $\bar{h}_2$, where $\bar{h}_2$ is the initial steady-state liquid level for the corresponding initial input $\bar{u}_2$. The following data set Z is collected from real time experiments under a uniform sample rate $t_s = t_{k+1} - t_k \equiv \text{constant}$, to identify the dynamics of the TT-TVS.

$$Z = \{h_1, h_2, u_1, u_2, t_s\} \quad (29)$$

By subtracting the initial steady-state liquid levels and the corresponding initial inputs by means of the relation given in (21), the deviation variables data set is represented as:

$$\hat{Z} = \{\hat{H}_1, \hat{H}_2, \hat{U}_1, \hat{U}_2, t_s\} \quad (30)$$

To identify the linear dynamics of TT-TVS, data set of deviation variables $\hat{H}_1(t)$ and $\hat{H}_2(t)$ are considered as outputs. Data set of deviation variables $\hat{U}_1(t)$ and $\hat{U}_2(t)$ are considered as inputs.

4. Linear dynamics identification of TT-TVS using system identification [6]

In the present section, system identification technique is employed to notify the linear model of TT-TVS [6]. A simple output error (OE) model structure is considered for the identification [6]. From (24), the linear dynamics of the TT-TVS can be expressed as follows:

$$\begin{bmatrix} \hat{H}_1(s) \\ \hat{H}_2(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \hat{U}_1(s) \\ \hat{U}_2(s) \end{bmatrix} \quad (31)$$

In (31), $G_{11}(s)$, $G_{12}(s)$, $G_{21}(s)$ and $G_{22}(s)$ are as follows:

$$G_{11}(s) = \frac{\hat{H}_1(s)}{\hat{U}_1(s) \mid \hat{U}_2(s)=0} \quad (32)$$

$$G_{12}(s) = \frac{\hat{H}_1(s)}{\hat{U}_2(s) \mid \hat{U}_1(s)=0} \quad (33)$$

$$G_{21}(s) = \frac{\hat{H}_2(s)}{\hat{U}_1(s) \mid \hat{U}_2(s)=0} \quad (34)$$

$$G_{22}(s) = \frac{\hat{H}_2(s)}{\hat{U}_2(s) \mid \hat{U}_1(s)=0} \quad (35)$$

To identify the linear s-domain model of TT-TVS, a data set of inputs ($\hat{U}_1$ and $\hat{U}_2$) and outputs ($\hat{H}_1$ and $\hat{H}_2$) of 5000 samples are collected with the sampling time of 0.1 s by running the real time TT-TVS. From the

experimental response (Fig. 2), it is observed that the response of water level in tanks is like a first order system response. Therefore, first order OE model is considered for TT-TVS model identification. Least square parameter estimation algorithm [6] is employed to estimate the parameters of the predicted model of TT-TVS. The identified linear transfer function model of TT-TVS is as follows:

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} = \begin{bmatrix} \frac{0.2862}{s+0.0103} & \frac{0.0316}{s+0.0136} \\ \frac{0.0354}{s+0.0131} & \frac{0.3516}{s+0.0134} \end{bmatrix} \quad (36)$$

The responses of identified model (36) and real time system are shown in Fig. 2.

From Fig. 2, it is notified that the identified model responses are almost tracking the real time responses.

5. Linear dynamics identification of TT-TVS using mathematical approach

In the present section, a new approach is identified to notify the linear dynamics of TT-TVS. The proposed identification procedure is as follows.

5.1. Identification of $G_{11}(s)$

$G_{11}(s)$ is the linear dynamics which are identified by taking the water height in Tank-1 as output variable and the voltage given to DC motor Pump-1 as input variable. As the response of water height in the Tank-1 is like first order system response, let the first order linear dynamics of $G_{11}(s)$ in the state space form as follows:

$$\dot{\hat{h}}_1(t) = a\hat{h}_1(t) + b\hat{u}_1(t) \quad (37)$$

Here $\hat{h}_1(t) = h_1(t) - \bar{h}_1$, $\hat{u}_1(t) = u_1(t) - \bar{u}_1$, $\bar{u}_1$ initial input and $\bar{h}_1$ is the initial steady-state water height for the corresponding $\bar{u}_1$. Parameters a and b of (37) are obtained by using real time water height response. it can be notified that $\hat{h}_1(t)$ is equal to zero at steady-state. At this condition, from (37), one can have

$$b = \frac{-a \times \hat{h}_{1ss}}{\hat{u}_1(t)} \quad (38)$$

where $\hat{h}_{1ss}$ is the steady-state water height of $\hat{h}_1(t)$. The solution of (37) shall be derived as follows:

$$\hat{h}_1(t) = \frac{b\hat{u}_1(t)}{a} [\exp(at) - 1] \quad (39)$$

By substituting (38) in (39), one obtains

$$\hat{h}_1(t) = \hat{h}_{1ss} [1 - \exp(at)] \quad (40)$$

At starting, $\bar{u}_1$ is considered as 2.85 V for 50 s. The corresponding initial steady-state water level is $\bar{h}_1 = 5.6$ cm. After 50 s, the input $u_1(t)$ is changed as:

$$u_1(t) = 2.85 + 0.45u(t - 50)V \quad (41)$$

where u is the unit-step signal. Which means that $\hat{u}_1(t) = 0.45V$. The water height $\hat{h}_1(t)$ for the corresponding $\hat{u}_1(t) = 0.45V$ is shown in Fig. 3.

From Fig. 3, it has notified that the steady-state water height in Tank-1 is $\hat{h}_{1ss} = 12.4$ cm. Using MATLAB software curve fitting technique, (40) is fitted with real time response ($\hat{h}_1(t)$) to obtain a and is found as $a = -0.0106$. By substituting $a = -0.0106$ in (38), it shall be found that $b = 0.2917$. Therefore, the identified linear state space model is as follows:

$$\dot{\hat{h}}_1(t) = -0.0106\hat{h}_1(t) + 0.2917\hat{u}_1(t) \quad (42)$$

By applying s-domain transformation to (42), it shall be obtained as

$$s\hat{H}_1(s) - \hat{h}_1(0) = -0.0106\hat{H}_1(s) + 0.2917\hat{U}_1(s) \quad (43)$$

By making initial value of $\hat{h}_1(0) = 0$, the linear transfer function dynamics of $G_{11}(s)$ are identified as:

$$G_{11}(s) = \frac{\hat{H}_1(s)}{\hat{U}_1(s)} = \frac{0.2917}{s + 0.0106} \quad (44)$$

Using the same procedure, $G_{12}(s)$, $G_{21}(s)$ and $G_{22}(s)$ are identified as follows:

$$G_{12}(s) = \frac{0.0462}{s + 0.0255} \quad (45)$$

$$G_{21}(s) = \frac{0.0499}{s + 0.0202} \quad (46)$$

$$G_{22}(s) = \frac{0.2457}{s + 0.0372} \quad (47)$$

Therefore, the identified linear dynamics of TT-TVS using the proposed method is as follows:

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} = \begin{bmatrix} \frac{0.2917}{s + 0.0106} & \frac{0.0312}{s + 0.0215} \\ \frac{0.0372}{s + 0.0201} & \frac{0.3294}{s + 0.0123} \end{bmatrix} \quad (48)$$

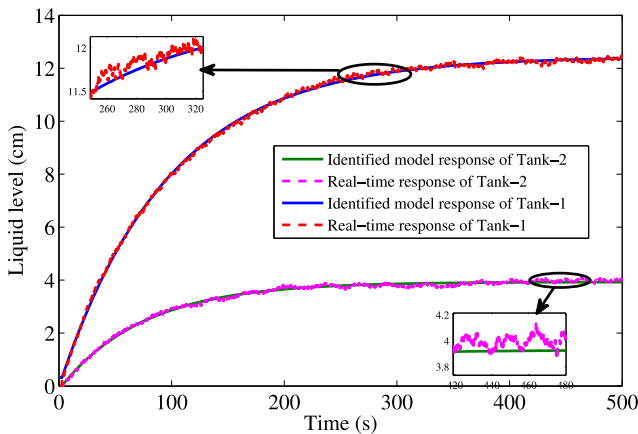


Fig. 2. Real time and notified dynamics outputs.

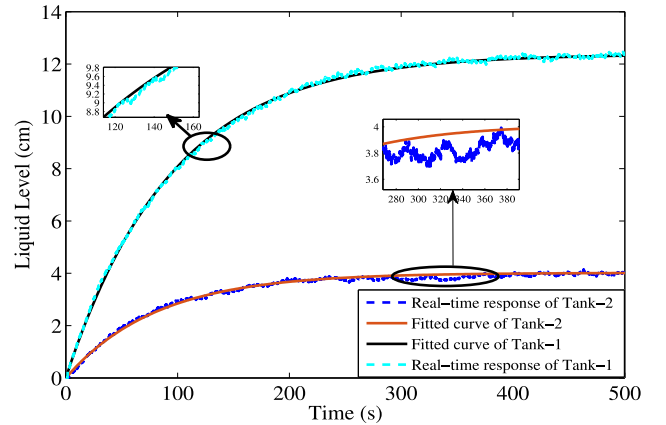


Fig. 3. Real time and curve fitted outputs.

Fig. 3 notifies that the identified dynamics response is very nearly tracking the real time experimental response.

6. Linear variable parameter transfer function (LVPTF) model identification of TT-TVS

In the present section, a linear variable parameter transfer function (LVPTF) model identification technique is presented to notify the linear dynamics of TT-TVS. The LVPTF model structure of TT-TVS is taken as follows:

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} = \begin{bmatrix} \frac{k_{11}(\hat{u}_1(t))}{\tau_{11}(\hat{h}_1(t))s + 1} & \frac{k_{12}(\hat{u}_2(t))}{\tau_{12}(\hat{h}_1(t))s + 1} \\ \frac{k_{21}(\hat{u}_1(t))}{\tau_{21}(\hat{h}_2(t))s + 1} & \frac{k_{22}(\hat{u}_2(t))}{\tau_{22}(\hat{h}_2(t))s + 1} \end{bmatrix} \quad (49)$$

The LVPTF modeling procedure is as follows.

6.1. Identification of $G_{11}(s)$

From (31) and (49), the LVPTF model of $G_{11}(s)$ is as follows:

$$G_{11}(s) = \frac{\hat{H}_1(s)}{\hat{U}_1(s)} = \frac{k_{11}(\hat{u}_1(t))}{\tau_{11}(\hat{h}_1(t))s + 1} \quad (50)$$

The LVPTF of $G_{11}(s)$ is notified from the real time run response. In order to notify the LVPTF, input variables data ($\hat{u}_1(t)$) and output variables data ($\hat{h}_1(t)$) are recorded. $k_{11}(\hat{u}_1(t))$ and $\tau_{11}(\hat{h}_1(t))$ are required to be identified in a way that the error margin between the output of (50) and real time output ($\hat{h}_1(t)$) to be as minimal as possible. Minimization can be performed in two ways. One way is optimization method or another way is curve fitting method. However, optimization methods effectiveness heavily relies on having a reasonably good initial guess for the parameter set and the defined search space. These algorithms operate by simulating the process of nature inspired evolution to optimize parameters, but the quality of the solution is often contingent on the starting conditions. Additionally, finding the global solution can be time-consuming, as it may require a large number of iterations to converge. Therefore, in the present work, required parameters k_{11} and τ_{11} are estimated by using curve fitting technique in MATLAB software and the finest fitted k_{11} is notified as:

$$k_{11}(\hat{u}_1(t)) = -0.0022\hat{u}_1^3(t) + 0.0237\hat{u}_1^2(t) - 0.0235\hat{u}_1(t) + 27.813 \quad (51)$$

The finest fitted τ_{11} is notified as:

$$\tau_{11}(\hat{h}_1(t)) = -0.005\hat{h}_1^3(t) + 0.0294\hat{h}_1^2(t) - 0.4603\hat{h}_1(t) + 94.8159 \quad (52)$$

Similarly, k_{12} , τ_{12} , k_{21} , τ_{21} , k_{22} and τ_{22} are found as:

$$k_{12}(\hat{u}_2(t)) = 0.0036\hat{u}_2^3(t) - 0.0075\hat{u}_2^2(t) + 0.0046\hat{u}_2(t) + 1.4504 \quad (53)$$

$$\tau_{12}(\hat{h}_1(t)) = -3.202\hat{h}_1^3(t) + 9.2313\hat{h}_1^2(t) - 7.1678\hat{h}_1(t) + 47.8695 \quad (54)$$

$$k_{21}(\hat{u}_1(t)) = 0.0047\hat{u}_1^3(t) - 0.009\hat{u}_1^2(t) + 0.0052\hat{u}_1(t) + 1.8498 \quad (55)$$

$$\tau_{21}(\hat{h}_2(t)) = -1.7736\hat{h}_2^3(t) + 6.8694\hat{h}_2^2(t) - 7.6064\hat{h}_2(t) + 51.3813 \quad (56)$$

$$k_{22}(\hat{u}_2(t)) = 0.0047\hat{u}_2^3(t) - 0.009\hat{u}_2^2(t) + 0.0052\hat{u}_2(t) + 1.8498 \quad (57)$$

$$\tau_{22}(\hat{h}_2(t)) = -0.006\hat{h}_2^3(t) + 0.0329\hat{h}_2^2(t) - 0.5268\hat{h}_2(t) + 82.3816 \quad (58)$$

From Fig. 4, it is notified that the identified model responses are almost tracking the real time responses. It is noted that factors such as eddy affects during liquid discharge, sloshing affect, hydraulic forces and turbulence in the water etc are non considerable while performing the modeling using conventional methods. These nonlinearities are challenging to model accurately using physical laws alone. Even if modeled experimentally, the overall model becomes highly nonlinear, making the design of controllers difficult. Therefore, there is a need for a designer-friendly model that can also account for the overall dynamics. The proposed LVPTF model identification technique is addressed these difficulties to a significant extent. However, the limitation of the proposed method is that it requires a steady outlet flow to satisfy the dynamics described in (1), which are derived based on Bernoulli's law. This steady flow assumption is crucial because Bernoulli's law assumes a consistent and unvarying flow rate to accurately describe the fluid dynamics.

From Figs. 2–4, it is observed that the identified models responses are almost tracking the real time responses. However, a numerical comparison among the identified models responses and real time responses is performed by taking integral absolute error (IAE) and root mean square error (RMSE) as performance indices [6]. The IAE and RMSE analysis is given Table 1. From the analysis in Table 1, comparatively, it is noticed that the LVPTF model identification technique provides better accuracy than the identification techniques in Refs. [6,16,17] and the identification technique in §V. In addition, the LVPTF model allows the designers to design adaptive controllers.

The International Federation of Automatic Control (IFAC) Committee defined a set of practical design problems to differentiate new and existing control methods and tools, enabling significant comparisons of control performance. The committee identified a series of real-world control issues, which were included as “benchmark control problems.” Among these, the level control problem in two tank-two variable system was highlighted as a benchmark in the category of MIMO/TITO control systems [6]. Control of liquid level in tanks and fluid flow between tanks is essential in nearly all process industries, such as wastewater treatment, chemical, petrochemical, pharmaceutical, food, and beverages. Effective level and flow control in tanks is crucial for the efficient operation of these industries. Due to MIMO characteristics TT-TVS poses challenging task to controller designers for liquid level control. To perform MIMO control, the proposed LVPTF model allows the designers to go for adaptive controllers design such as adaptive PID family controllers, adaptive fractional order PID family controllers and adaptive dual mode PID family controllers.

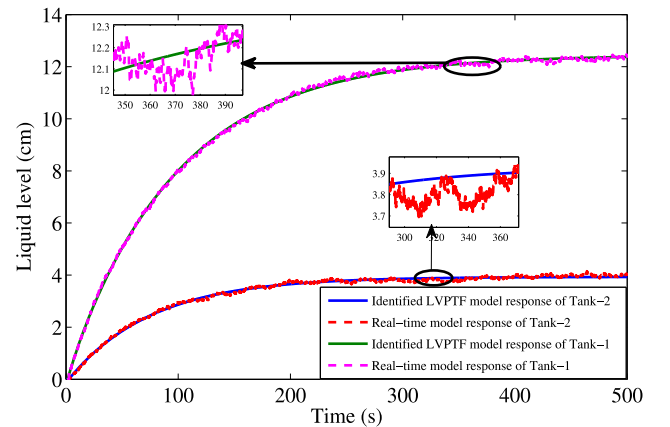


Fig. 4. Real time and notified LVPTF outputs.

Table 1
Performance analysis of identification methods.

Identification method	IAE in cm		RMSE in cm	
	for $G_{11}(s)$	for $G_{22}(s)$	for $G_{11}(s)$	for $G_{22}(s)$
Method in §IV [6]	0.6142	0.8451	0.5124	0.6254
Identification method in [16]	0.5961	0.7451	0.6124	0.6130
Identification method in Ref. [17]	0.4814	0.6756	0.6621	0.6534
Proposed method in §V	0.5204	0.7021	0.4256	0.5613
Proposed method in §VI	0.4818	0.6513	0.3425	0.3524

7. Conclusion

In this paper, two novel linear modeling techniques are proposed for TT-TVS dynamics identification. The identification is performed using the real time input-output data. Linearization of nonlinear dynamics of TT-TVS using Taylor series expansion is described. A numerical comparison is presented among the proposed identification techniques and existing literature identification approaches. From the comparison, it is observed that the LVPTF model provides better accuracy in TT-TVS linear dynamics identification and it allows the designer to design adaptive controllers for liquid level regulation in the TT-TVS. The main advantages of the proposed LVPTF method over the conventional methods is that the un-modeled affects such as eddy affects during liquid discharge, sloshing affect, hydraulic forces and turbulence in the water etc are well addressed and user friendly model can be obtained in the above mentioned difficult circumstances. As future scope, the proposed LVPTF model can be used to design adaptive linear controllers such as adaptive PID family controllers, adaptive fractional order PID family controllers and adaptive dual mode PID family controllers.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Sreekanth Reddy Arava reports was provided by Rajiv Gandhi University of Knowledge Technologies. Sreekanth Reddy Arava reports a relationship with Rajiv Gandhi University of Knowledge Technologies that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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